MAE 3240 Problem Set #2

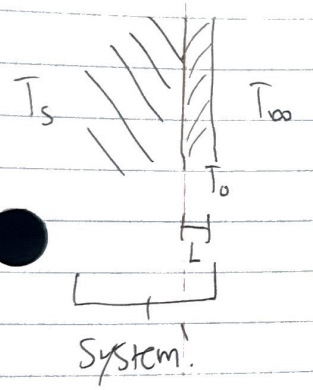
Gregory Sharma

#1 $R = 0.5$, $H = 1$, $T_s = 50^\circ C$, $T_\infty = 20^\circ C$, $h = 15 \frac{W}{m^2 K}$

a. $A = [(2\pi R)H + \pi R^2]$, $q_{in} = hA(T_s - T_\infty) \rightarrow q = (15[2\pi(0.5)(1) + \pi(0.5)^2] \times (50-20)$

$$\boxed{q = \sim 1767\ W}$$

b. $L = 0.02m$, $k = 0.1 \frac{W}{mK}$



$q_{in} = q_{out}$] insulation. $\rightarrow$ $kA\left(\frac{T_s - T_0}{L}\right) = hA(T_0 - T_\infty)$

$q_{in} = q_{out}$] System! $\frac{0.1}{0.01}(50 - T_0) = 15T_0 - 300$

$T_0 = 27.5^\circ C$

$q_{in} = hA(T_0 - T_\infty) = (15)(2\pi(0.5)(1) + \pi(0.5)^2)(27.5 - 20)$

$$\boxed{q_{in} = 442\ W}$$

c. By adding thermal insulation, the power needed to maintain the water temp is much reduced (by a factor of ~4). This is because it reduces the rate of heat transfer that can flow to the surface thus reducing the external temp. achieved at S.S which reduces the rate of heat removed by convection!

Gregory Shalma

293.15 K

(2)(a) $L = 0.1\,m$, $k = 1 \frac{W}{mK}$, $T_\infty = 20°C$, $h = 100 \frac{W}{m^2 K}$

$q'' = 1000 \frac{W}{m^2}$ $\quad q''_x = -k \frac{dT}{dx}$, $q'' = h(T_s - T_\infty)$

$h(T_s - T_\infty) = 1000$, $100 T_s - 100(293.15) = 1000 \rightarrow T_s = 303.15\,K$

$q''_x = k\left(\frac{T_0 - T_s}{L}\right) \rightarrow 1000 = \underbrace{k}_{=1} \frac{(T_0 - 303.15)}{0.1} \rightarrow T_0 = 403.15\,K = \boxed{130°C}$ a.

$\cancel{q''_{max} = k\left(\frac{200 - T_s}{0.1}\right) \rightarrow q''_{max} = 2000 - 10T_s}$

$\cancel{q''_{max} = h(T_s - T_\infty) \rightarrow q''_{max} = 100 T_s -}$

(2b)

$200°C = 473.15\,K$

$q''_{max} = k\left(\frac{473.15 - T_s}{0.1}\right) \rightarrow q''_{max} = 4731.5 - 10T_s$

$q''_{max} = h(T_s - T_\infty) \rightarrow q''_{max} = 100(T_s - 293.15) = 100T_s - 29315$

$4731.5 - 10T_s = 100T_s - 29315 \rightarrow T_s = 309.5\,K$

$q''_{max} = 4731.5 - 10(309.5)$: $\boxed{q''_{max} = 1636.5 \frac{W}{m^2}}$

(2c)

$q''_{max} = k\left(\frac{T_{max} - T_s}{L}\right)$ $\quad q''_{max} = h(T_s - T_\infty)$

$k\uparrow$ $\quad L\downarrow$ $\quad h\uparrow$

(1) Could increase the thermal conductivity "k" of the material to draw more heat out faster

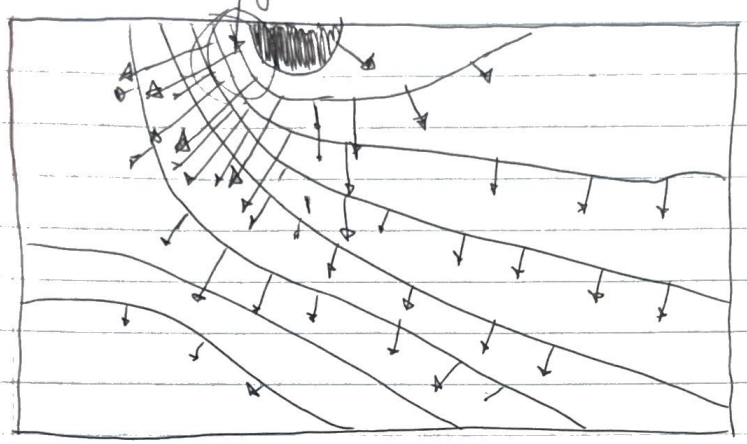
(2) Could decrease "L", the length of the film

(3). Could increase the heat trans. Coef "h" of the coolant to draw more heat

Gregory Sharma

(#3) are perpendicular

(a) greater here!



(b.) The left side has the greater heat flux because the heat flux vector are place more dencho/are with greater magnitude. There is a large change in temperature across a shorter distance in the same medium which also indicates a larger heat flux!